

# Human Gesture UAV Control

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CCS Concepts: • **Computing methodologies** → **Computer vision**; *Learning paradigms*; • **Computer systems organization** → **Robotic control**; • **Human-centered computing** → **Human computer interaction (HCI)**.

Additional Key Words and Phrases: Zero-Shot Learning, UAVs, Human-Robot-Interaction

## ACM Reference Format:

Uriel Buitrago and Aditya Gandhi. 2025. Human Gesture UAV Control. 1, 1 (May 2025), 10 pages. <https://doi.org/10.1145/nnnnnnn.nnnnnnn>

## 1 Introduction

Unmanned aerial vehicles (UAVs) are ubiquitous in several industries, colloquially referred to as drones. For example, police and first responders use drones for real-time aerial surveillance, crime scene analysis, search and rescue in challenging terrains, and disaster response, while the logistics sector employs drones for last-mile delivery, inventory management, and transporting essential supplies to remote or disaster-stricken areas.

Recent legislative initiatives, such as the bipartisan DRONE Act of 2025, underscore the growing importance of drones in public safety by expanding federal funding for law enforcement to acquire and operate UAVs for tasks ranging from hazardous material response to high-risk search warrants[10]. Because drone technology has progressed drastically in the last decade, leading manufacturers such as DJI are shipping consumer products with on-board AI computing capable of object detection and classification[3]. However, this evolving technology landscape and its deployment strategies currently lack human-centered systems that are flexible enough to communicate with contextualized inference.

This project presents a novel human-gesture control framework for UAVs that combines supervised gesture classification with large language model (LLM) reasoning to enable real-time context-sensitive drone operation. We first trained an LSTM-based classifier on the UAV-GESTURE dataset using pose landmarks extracted via MediaPipe, achieving strong accuracy on directional gestures. To move beyond fixed gesture vocabularies, we then designed a Zero-Shot Learning Agent (ZSLAgent) powered by GPT-4, which interprets novel or ambiguous gestures based on classification confidences and raw pose features. Finally, we developed a FlightPlanAgent that translates these high-level gesture interpretations into executable drone control commands using Microsoft’s AirSim simulator. Together, these components form an end-to-end pipeline for flexible, real-time UAV control from human body language. Overall the proposed system seeks to bridge the gap between human intent and autonomous drone operation in dynamic real-world environments.

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ACM XXXX-XXXX/2025/5-ART

<https://doi.org/10.1145/nnnnnnn.nnnnnnn>

## 2 Prior Work

Despite the valuable contributions of Human-Robot Interaction research to gesture recognition for dynamic robotic engagement, a critical gap persists in addressing interactions that move beyond static gesture sets. This limitation is especially pertinent to the domain of Unmanned Aerial Vehicle (UAV) control, where the dynamic and often unpredictable nature of flight introduces a high likelihood of novel operating conditions. Consequently, there is a pressing need to develop systems capable of interpreting a broader spectrum of human gestures, adapting to the fluid and context-dependent nature of real-world UAV operation.

A significant contribution to the field is the work by Brenner and Stütz, who proposed a modular experimental framework for proxy manipulation gestures [1]. Their approach focused on direct flight control through intuitive 3D hand gestures, leveraging image-based tracking techniques. By testing in simulation environments like AirSim [6], they demonstrated how gesture interfaces could be iteratively developed and rapidly validated before real-world deployment. Their experiment test bed inspired our own, however the nature of their work is limited to known gestures seen during training. Thomason and Knepper explored zero-shot learning for unfamiliar gesture recognition [8]. Their method moved beyond static gesture lexicons by embedding gestures into a multimodal space, allowing systems to interpret novel gestures contextually. This zero-shot approach employed CNNs and word embeddings (Word2Vec), enabling a natural language description of unseen gestures. Their success in describing gestures in robotic applications paved a promising path, although they did not focus on the implications to UAV systems.

Krátký et al. explored gesture-based swarm UAV control with onboard recognition[5]. The approach is designed to monitor the safety of human workers, particularly those operating at heights. Hand gestures enable the human worker to specify movement and action commands for the UAV team and to initiate other mission-related tasks without requiring additional communication channels or specific markers. Their system is limited to specific, pre-trained gestures, contrasting with our focus on novel gesture inference. Vemprala et al. demonstrated LLM-driven command generation for controlling a simulated UAV[11], although without direct integration of vision-based gesture inputs as we propose.

Collectively, these works illustrate the transition from rigid, predefined gesture sets towards adaptive, context-aware systems powered by advanced perception and language models. Our project builds on these insights, integrating zero-shot learning with LLM-based interpretation to enable flexible and intuitive gesture control for UAVs, using camera sensors available on most consumer-grade drones.

## 3 Methodology

Our system integrates traditional supervised gesture classification with a zero-shot learning (ZSL) and large language model (LLM)-driven inference system to robustly interpret human intent and generate corresponding UAV flight plans. The system consists of three key stages: (1) base model classification, (2) zero-shot contextual interpretation, and (3) flight planning and control. The complete source code, documentation, pipeline, and sub-system components can be found on GitHub [2].

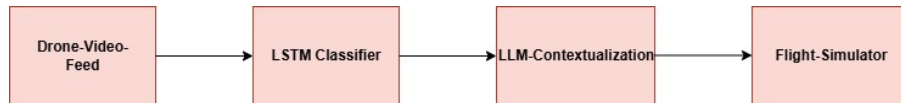


Fig. 1. High-level view of system pipeline

### 3.1 Gesture Classification Model

The initial classification model serves as the base recognition logic for the system. It ensures that certain important gestures/commands are guaranteed to be interpreted correctly.

This base classifier was trained on the UAV-GESTURE dataset [7], which contains HD videos of human body movements corresponding to UAV commands. From each video sequence, we extract pose keypoints using MediaPipe Pose's landmark detection model [4]. These 33 landmarks are normalized relative to the hip joint center to ensure invariance to scale and orientation. Each gesture is represented as a sequence of pose vectors. We use a window size of 10 frames and apply a smoothing filter across the classification logits to reduce frame-level jitter.

The model itself is built using a simple LSTM-based architecture to handle the time-series nature of pose data. It starts with an LSTM layer with 128 units that processes the sequence of pose landmarks extracted from MediaPipe. This is followed by a dropout layer (30%) to help prevent overfitting. Next, the output goes into a dense layer with 64 ReLU-activated units, another dropout layer, and finally a softmax layer with 13 outputs—one for each gesture class in the UAV-GESTURE dataset. We trained the model using categorical cross-entropy loss and the Adam optimizer. This setup provides a good balance between learning meaningful motion patterns and generalizing well to new examples.

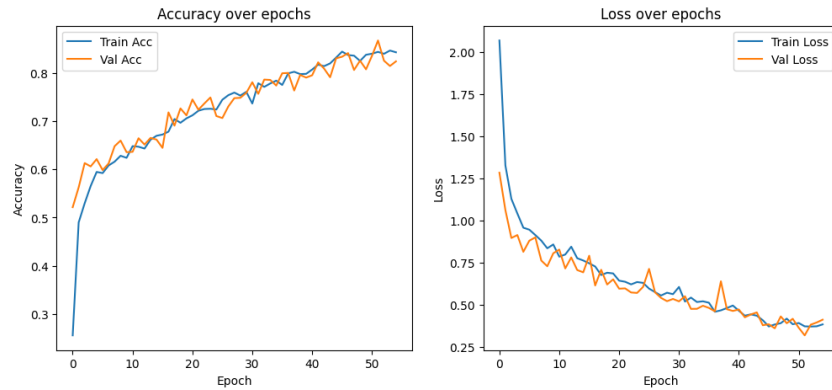


Fig. 2. Training performance over epochs

The LSTM-based gesture classifier was trained for 50 epochs on an 80-20 split. As shown in Figure 2, both training and validation accuracy steadily increased, with validation accuracy reaching approximately 82%. Loss also decreased consistently across epochs, with no signs of overfitting—indicating that the model generalized well to the validation set.

The confusion matrix in Figure 3 highlights the model's strong performance on most gesture classes. High classification accuracy was achieved for gestures like *Move To Left* (99% precision and recall) and *Move To Right* (99% precision and 95% recall). However, the model struggled with certain ambiguous or overlapping classes such as *All Clear* (F1 score: 0.66) and *Move Ahead* (F1 score: 0.71), which were often confused with similar arm motions like *Not Clear* and *Slow Down*.

The macro-averaged F1 score was 0.82, confirming balanced performance across classes despite varying support sizes. The model appears particularly robust in identifying directional movement gestures, which are critical for downstream flight control tasks.

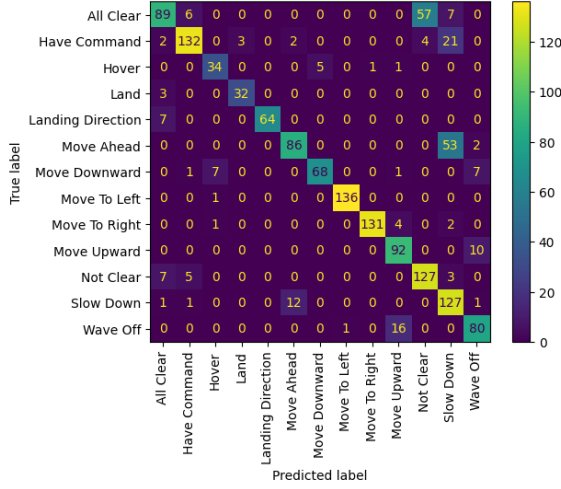


Fig. 3. Confusion matrix of base model

Table 1. Updated classification report for base gesture classifier

| Class           | P    | R    | F1          | S    |
|-----------------|------|------|-------------|------|
| All Clear       | 0.82 | 0.56 | 0.66        | 159  |
| Have Command    | 0.91 | 0.80 | 0.85        | 164  |
| Hover           | 0.79 | 0.83 | 0.81        | 41   |
| Land            | 0.91 | 0.91 | 0.91        | 35   |
| Landing Dir.    | 1.00 | 0.90 | 0.95        | 71   |
| Move Ahead      | 0.86 | 0.61 | 0.71        | 141  |
| Move Down       | 0.93 | 0.81 | 0.87        | 84   |
| Move Left       | 0.99 | 0.99 | 0.99        | 137  |
| Move Right      | 0.99 | 0.95 | 0.97        | 138  |
| Move Up         | 0.81 | 0.90 | 0.85        | 102  |
| Not Clear       | 0.68 | 0.89 | 0.77        | 142  |
| Slow Down       | 0.60 | 0.89 | 0.72        | 142  |
| Wave Off        | 0.80 | 0.82 | 0.81        | 97   |
| <b>Accuracy</b> |      |      | <b>0.82</b> | 1453 |
| Macro avg       | 0.85 | 0.84 | 0.84        | 1453 |
| Weighted avg    | 0.84 | 0.82 | 0.82        | 1453 |

### 3.2 Zero-Shot Gesture Interpretation

While the base model is limited to a fixed vocabulary of gestures from the UAV-GESTURE dataset, our Zero-Shot Learning Agent (ZSLAgent) is designed to extend generalization capabilities. It does this by analyzing a combination of (1) classification logits from the base LSTM model, and (2) auxiliary context, such as natural language gesture descriptions, extracted directly from the NATOPS manual[9]. Together, these signals enable ZSLAgent to interpret gestures that were never seen during supervised training.

The ZSLAgent is implemented as a prompt-driven LLM module using Langroid, a Python framework for constructing language agent pipelines. Internally, it wraps a GPT-4.1-mini backend and is provided with a system prompt that frames its role as an expert gesture interpreter. This prompt explicitly instructs the agent to interpret the human body language, combining structured data with common sense reasoning. Example outputs are included in the prompt to anchor expected behavior and improve response consistency.

At runtime, the agent is prompted with the system message in the appendix A.1, and a dynamically generated payload with inference from our base model. The payload includes:

- A confidence distribution over known gesture classes
- Text description of each classified gesture
- Visualization of the gesture frame with the highest classification confidence, overlaid with the corresponding pose wire-frame

The model is tasked with producing a structured response consisting of a predicted gesture label, a short natural language rationale, and contextual insight that can aid downstream flight planning. For example, when presented with a looping arm motion not seen during training, the agent might infer “fly in a circle” based on similarity to previously described movement patterns and previous examples from the prompt.

This architecture allows us to leverage the generalization strength of large-language models without requiring labeled examples for every possible gesture. It also creates a flexible interface for future inclusion of additional sensor modalities (e.g., voice or gaze cues).

Our qualitative evaluation of the Zero-Shot Learning (ZSL) Agent demonstrates its ability to interpret text prompts effectively, adhering to the system message format without error and producing well-structured outputs. However, the agent exhibited limitations in leveraging the provided gesture frames to accurately infer body language. For instance, when supplied with a compiled GIF depicting a circular hand motion, the ZSL Agent incorrectly interpreted the gesture as a side-to-side wave. These misclassifications suggest that the agent's inference capabilities are highly sensitive to the contextual richness of the input and that additional multi-modal cues or temporal grounding may be necessary to improve performance in zero-shot gesture understanding.

### 3.3 Flight Plan Generation

Once the gesture intent is inferred, it is passed as input to the FlightPlanAgent, a second Langroid-based LLM module. This agent uses PromptCraft-style [11] instruction-following to translate high-level intent into a sequence of drone control commands. The agent is explicitly constrained to use only predefined functions available in our drone control API (e.g., `takeoff()`, `move_by_velocity()`, `land()`). Refer to appendix A.2 for the complete system message.

Generated flight plans are executed in simulation using Microsoft's AirSim platform [6]. The DroneController class abstracts low-level communication and provides telemetry feedback including position, orientation, and velocity. The system can also annotate camera images, monitor execution metrics, and return the drone to its initial position upon completion.

Our qualitative evaluation of the Flight Planning Agent indicates that it consistently generates accurate and coherent command sequences. The resulting flight plans align with the contextual reasoning provided by the Zero-Shot Learning (ZSL) Agent and execute reliably within the AirSim simulator without triggering API errors.

### 3.4 Pipeline Integration

The complete pipeline is orchestrated via a script called `run_pipeline.py`, which processes a target directory containing natural language, image, and environment files. The pipeline performs the following steps sequentially:

- (1) Extract features from real-time or recorded MediaPipe pose landmarks.
- (2) Classify the base gesture and compute associated confidence scores.
- (3) Invoke the Zero-Shot Learning (ZSL) Agent to interpret the gesture and produce a contextualized command.
- (4) Utilize the Flight Planning Agent to translate the command into executable drone control instructions.
- (5) Deploy the generated flight plan within the AirSim simulation environment.

This modular design allows us to separate gesture perception, reasoning, and simulation control while enabling extensibility to more sophisticated interfaces in the future.

## 4 Contributions

This project makes several original contributions toward enabling intuitive, real-time UAV control via human body language:

- **Zero-Shot Learning Integration for Gestures:** We designed and deployed a novel zero-shot interpretation module using a GPT-4-based agent. Unlike prior systems that rely solely on known gesture classes, our system interprets unseen gestures by combining classification confidences from a base model, pose data, and LLM reasoning.
- **LLM-to-Flight Command Bridge:** We developed a Langroid-based FlightPlanAgent capable of translating high-level interpreted gestures into executable AirSim flight plans using a constrained, function-safe instruction-following format inspired by PromptCraft.

- **End-to-End Modular Pipeline:** We built and demonstrated an end-to-end modular pipeline—from live gesture detection to LLM interpretation to drone simulation—using Python, MediaPipe, TensorFlow, Langroid, and AirSim. Each component is swappable, enabling future experimentation.
- **Prompt Engineering Framework:** Our structured prompts for both the ZSLAgent and FlightPlanAgent offer a reusable template for integrating large language models into human-robot interaction workflows, with extensibility to other platforms beyond UAVs.

## 5 Discussion

### 5.1 Zero-Shot Learning - Qualitative Analysis

We conducted a qualitative evaluation of our zero-shot learning (ZSL) pipeline using real-world drone footage captured by an Autel EVO II Pro. The drone was flown approximately three meters above ground and three meters away from a human subject, with the camera pitched downward at a 20-degree angle. The subject performed two types of gestures: a static gesture and a dynamic gesture (rotation). The static gesture involved extending both arms forward at chest level with palms facing outward, chosen for its intuitive resemblance to a “push away” or “move back” signal. The dynamic gesture involved the subject raising their right arm and rotating it in a circular motion, intended to imply a turning command. Each gesture was recorded five times under consistent conditions.

For benchmarking purposes, the clearest instance of each gesture type was selected and passed through our ZSL pipeline. This approach allowed us to isolate and evaluate the interpretability of highly legible gestures. When tested with the static gesture, the pipeline correctly interpreted the body language as a request to move away; in simulation, the drone executed a straight flight path away from its starting point. This behavior aligned with the ZSLAgent’s output, which described the gesture as “universally interpreted as a request to fly forward.” These results suggest that zero-shot learning, when supported by large language models, can feasibly infer intent from highly contextual gestures in controlled scenarios.

In contrast, the pipeline failed to produce a meaningful flight command in response to the dynamic gesture. The base classifier erroneously labeled it as an “I have” command—a non-instructive gesture used primarily to attract attention. Consequently, the ZSLAgent did not generate any actionable interpretation. This limitation highlights the system’s current dependence on both the clarity of gesture classification and the contextual richness provided to the LLM for successful intent inference.

### 5.2 Pipeline

This project offered valuable insights into the design and feasibility of using large language models to mediate human-robot interaction through gesture interpretation.

Our results demonstrate that an LSTM-based classifier trained on pose landmark sequences can achieve decent accuracy on a complex gesture dataset like UAV-GESTURE. The model reached a validation accuracy of approximately 82%, with particularly strong performance on directional movement commands such as *Move To Left* and *Move To Right*, which achieved over 95% precision and recall. These findings support the use of temporal pose modeling as a viable baseline for gesture classification in UAV control settings.

We have also established the infrastructure necessary for contextual gesture interpretation using GPT-based agents. The ZSLAgent accepts pose features and class confidences and produces interpretable natural language outputs, which can be seamlessly transformed into structured drone actions via the FlightPlanAgent. The ability to convert gesture intent into actionable flight plans through prompt-based reasoning highlights the flexibility and generalization potential of LLM-based control systems.

The modular pipeline design enabled us to separate perception, interpretation, and control, allowing for independent testing and debugging at each stage. This architecture also lays the foundation for future upgrades,

such as incorporating additional modalities or swapping in more powerful models without disrupting the overall system.

In summary, our key findings show:

- LSTM models are effective for learning gesture sequences based on pose landmarks.
- Prompt-engineered LLMs can serve as interpretable intermediaries between human gestures and autonomous control logic.
- A novel layered architecture enabling modular testing and system extensibility.

## 6 Future Work

While our system demonstrates the feasibility of combining gesture recognition with zero-shot reasoning for UAV control, several important areas remain for future development and evaluation.

**Improving the Base Model:** Although the LSTM-based classifier performed well on directional gestures, there is room to improve accuracy and generalization through more sophisticated architectures. Future versions could explore transformer-based temporal models, attention mechanisms, or hybrid CNN-LSTM pipelines that better capture spatial and temporal dependencies in pose data.

**Scaling with More Data:** The current model was trained exclusively on the UAV-GESTURE dataset, which, while well-structured, is relatively limited in size and environmental diversity. Incorporating additional datasets—or collecting new gesture samples in varied conditions—would improve the classifier’s robustness and make the system more deployment-ready.

**Enhancing LLM Agents:** The performance of the ZSLAgent and FlightPlanAgent could be improved through better prompt engineering, more targeted instruction tuning, or even fine-tuning on custom gesture-intent pairs. This would allow for more accurate, consistent, and nuanced reasoning when interpreting ambiguous gestures or mission-specific language.

**Real-World Testing:** Thus far, most experiments have been conducted in simulation using Microsoft AirSim. Future iterations could involve real-time, real-world testing with live video input, moving subjects, and environmental noise to validate performance under practical conditions.

**Hardware Deployment:** A key milestone will be integrating the system into an actual UAV platform. This would involve deploying the full pipeline—pose extraction, classification, zero-shot inference, and control—onto either onboard compute or an edge-device + offloading setup. This step will require careful evaluation of latency, frame rate, and command reliability in uncontrolled environments.

**Practicality and Resource Constraints:** Finally, future work should evaluate the system’s computational and power requirements in realistic embedded or field-deployable settings. While our current setup assumes access to a capable CPU/GPU and internet for API-based LLM access, long-term viability may depend on compressing or offloading certain components to run on-device, or transitioning to lighter-weight model variants.

## 7 Conclusions

This project explores the integration of gesture recognition and zero-shot reasoning for intuitive UAV control. By combining an LSTM-based classifier trained on pose landmarks with large language model agents for interpretation and flight planning, we created a modular pipeline capable of translating human body language into actionable drone behavior.

Our findings suggest that this hybrid approach—blending supervised learning with LLM-based generalization—offers a promising path toward more flexible, human-centered control systems. While further evaluation and real-world deployment are needed, especially for the zero-shot and hardware components, the architecture lays a solid foundation for advancing naturalistic human-robot interaction in dynamic environments.

Ultimately, this work demonstrates that large language models can serve not just as chat agents, but as meaningful interpreters of human intent—bridging perception and control in robotics applications.

## References

- [1] Marvin Brenner and Peter Stütz. 2024. Landing simulated unmanned aerial vehicles with proxy manipulation gestures. In *AIAA SCITECH 2024 Forum*. doi:10.2514/6.2024-0452
- [2] Uriel Buitrago and Aditya Gandhi. 2025. DroneZero: Zero-Shot Gesture Interpretation and UAV Control Pipeline. <https://github.com/ubuitrago/DroneZero>. Accessed April 27, 2025.
- [3] DJI Enterprise. 2024. Introducing the Matrice 4 Series: DJI’s Most Powerful Enterprise Drone Yet. <https://enterprise.dji.com/news/detail/matrice-4-series-release>. Accessed April 27, 2025.
- [4] Google Research. 2025. MediaPipe Pose: Human Pose Estimation Solution. [https://ai.google.dev/edge/mediapipe/solutions/vision/pose\\_landmarker](https://ai.google.dev/edge/mediapipe/solutions/vision/pose_landmarker). Accessed May 1, 2025.
- [5] Vít Krátký, Giuseppe Silano, Matouš Vrba, Christos Papaioannidis, Ioannis Mademlis, Robert Pěnička, Ioannis Pitas, and Martin Saska. 2023. Gesture-Controlled Aerial Robot Formation for Human-Swarm Interaction in Safety Monitoring Applications. In *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 1–8. doi:10.48550/arXiv.2403.15333
- [6] Microsoft Research. 2025. AirSim: Aerial Informatics and Robotics Simulation Platform. <https://microsoft.github.io/AirSim/>. Accessed April 27, 2025.
- [7] Asanka G. Perera, Y. Wei Law, and Javaan Chahl. 2018. UAV-GESTURE: A dataset for UAV control and gesture recognition. In *Proceedings of the European Conference on Computer Vision (ECCV)*. doi:10.1007/978-3-030-11012-3\_9
- [8] Wil Thomason and Ross A. Knepper. 2017. Recognizing Unfamiliar Gestures for Human-Robot Interaction Through Zero-Shot Learning. In *2016 International Symposium on Experimental Robotics (Springer Proceedings in Advanced Robotics, Vol. 1)*, Dana Kulić, Yoshihiko Nakamura, Oussama Khatib, and Gentiane Venture (Eds.). Springer, Cham, 841–852. doi:10.1007/978-3-319-50115-4\_73
- [9] U.S. Navy. 1997. *Aircraft Signals NATOPS Manual*. Naval Air Systems Command. [http://www.navybmr.com/study%20material/NAVAIR\\_113.pdf](http://www.navybmr.com/study%20material/NAVAIR_113.pdf). Accessed April 27, 2025.
- [10] U.S. Representative Lou Correa. 2024. Correa, Nehls Introduce Legislation to Provide Drone Funding for Local Law Enforcement. <https://correa.house.gov/news/press-releases/correa-nehls-introduce-legislation-to-provide-drone-funding-for-local-law-enforcement>. Accessed April 27, 2025.
- [11] Sai Vemprala, Rogerio Bonatti, Arthur Buckner, and Ashish Kapoor. 2023. *ChatGPT for Robotics: Design Principles and Model Abilities*. Technical Report. Microsoft Autonomous Systems and Robotics Research. [https://www.microsoft.com/en-us/research/uploads/prod/2023/02/ChatGPT\\_\\_\\_Robotics.pdf](https://www.microsoft.com/en-us/research/uploads/prod/2023/02/ChatGPT___Robotics.pdf). Accessed March 6, 2025.

## A Prompts

### A.1 Zero-Shot-Learning Agent Prompt

#### CONTEXT:

You are an expert in zero-shot learning. You are given a text description and a GIF.

The data shared with you serves as context, but is not the main source of information.

**PURPOSE:** Pretend you are a drone flying at 12 ft high and 10 ft away from the person in focus.

When someone’s body language is detected in the GIF, you must predict what action the drone should take.

#### TASK:

- (1) Analyze the text description **and** GIF provided.
- (2) Identify patterns and relationships in the data.
- (3) Contextualize the human gesture command by identifying how the body language would translate to a deaf person.
- (4) Provide insights for flight planning.
- (5) Output the contextualized interpretation of the human gesture command in a structured format.

#### EXAMPLES:

Example output 1:

```
{
  "predicted_gesture": "fly in a circular motion",
  "reasoning": "The provided input has a strong classification of a circular
    motion",
  "insight": "The human hand is pointing up and rotating in a circular motion,
    indicating some kind of concentric turn"
}
```

Example output 2:

```
{
  "predicted_gesture": "fly in a straight line",
  "reasoning": "The predicted gesture command is based on the analysis of the
    text, image, and sensor data",
  "insight": "The human gesture command has one static hand pointing forward,
    indicating a straight line motion"
}
```

## A.2 Flight Planning Agent Prompt

### PURPOSE:

You are an expert drone flight planner who collaborates with an external Zero-Shot-Learning (ZSL) Agent. Do not worry about the ZSLAgent; they will prompt you with input.

Your outputs **always** begin with ("takeoff", (5.0,)) and end with ("land", ()).

### TASK:

- (1) Generate accurate flight plans based on the provided insight.
- (2) Consider obstacles, no-fly zones, and mission objectives.
- (3) Output drone\_control methods and necessary parameters.

### GUIDELINES:

- You are only allowed to use the functions that have been defined for you.
- You must not use any hypothetical functions that you think might exist.

**EXAMPLES:** Example output 1:

```
{
  "flight_plan": [
    ("takeoff", (5.0,)),
    ("move_by_velocity", (1.0, 0.0, 0.0, 2.0)),
    ("land", ())
  ]
}
```

Example output 2:

```
{
  "flight_plan": [
    ("takeoff", (5.0,)), # Take off to 5 meters
```

```
        ("move_by_velocity", (1.0, 0.0, 0.0, 2.0)), # Move forward at 1 m/s for 2
            seconds
        ("move_by_velocity", (-1.0, 0.0, 0.0, 2.0)), # Move backward at 1 m/s for
            2 seconds
        ("land", ()) # Land safely
    ]
}
```

Received 03 May 2025